# BEVS Numericals — Formula Sheet

Study-ready formulas extracted from *Numericals – 1: Revised Solutions*

**Scope and notation:** Based on the available 11-page revised PDF prepared from a 26-page handwritten scan. Page references below refer to the available revised PDF. Chemical subscripts/superscripts are restored where text extraction corrupted them; no unreadable numerical inputs have been guessed.

## 1. Gas Volume and Air-Pollutant Concentrations

### 1.1 Ideal-gas molar volume

**V■ = V■ × (T■/T■)**
Finds gas volume at a new temperature when pressure and gas amount are constant.
**Symbols:** V■, V■ = initial/final volume; T■, T■ = absolute temperature.
**Conditions:** Temperatures must be in kelvin. At 0 °C and 1 atm, the source uses 22.4 L/mol.
**PDF ref:** p. 2

**T(K) ≈ T(°C) + 273**
Converts Celsius to kelvin.
**Symbols:** T = temperature.
**Conditions:** Use kelvin in the volume relationship.
**PDF ref:** p. 2

**Examples:** At 25 °C: 22.4 × 298/273 ≈ 24.5 L/mol. At 100 °C: 22.4 × 373/273 ≈ 30.6 L/mol.

> **Note:** The handwritten 100 °C value appears as 104.5 L/mol, but the revised PDF flags it as inconsistent; approximately 30.6 L/mol follows the stated relationship at 1 atm.

### 1.2 Gas concentration to ppm by volume at STP

**ppmv = [C (mg/m³) × 22.4]/MW**
Converts gas concentration from mg/m³ to ppm by volume.
**Symbols:** C = concentration in mg/m³; MW = molecular weight in g/mol; 22.4 = molar volume in L/mol at STP.
**Conditions:** Convert µg/m³ to mg/m³ first; use the STP convention in the source.
**PDF ref:** p. 2

**Conversions/examples:** 1,000 µg = 1 mg. CO: 30 mg/m³, MW 28 → 24.0 ppmv. NO■: 400 µg/m³ = 0.400 mg/m³, MW 46 → ≈0.195 ppmv. SO■: 350 µg/m³ = 0.350 mg/m³, MW 64 → ≈0.123 ppmv.

### 1.3 Annual vehicle emissions

**Annual distance = number of vehicles × average distance per vehicle per year**
Calculates total annual fleet distance.
**Symbols:** Distance typically in km/vehicle/year.
**Conditions:** Use values and units stated in the problem.
**PDF ref:** p. 2

**Annual pollutant mass = annual distance × emission factor**
Calculates annual emissions from a distance-based factor.
**Symbols:** Emission factor may be in g/km.
**Conditions:** If the factor is g/km, result is in grams before conversion.
**PDF ref:** p. 2

**Tonnes = mass (g)/10■**
Converts grams to metric tonnes.
**Symbols:** 1 tonne = 1,000 kg = 10■ g.
**Conditions:** Apply when calculated mass is in grams.
**PDF ref:** p. 2

For solvent/paint examples, total annual use = number of users/vehicles × annual solvent/paint use per unit (p. 2).

## 2. Thermal Power Plant and Fuel Combustion

### 2.1 Annual generation and fuel use

**Annual electricity output (MWh) = plant capacity (MW) × 8,760 h/year × load factor**
Estimates annual generation.
**Symbols:** Capacity in MW; load factor as a fraction (70% = 0.70).
**Conditions:** 8,760 h assumes a 365-day year.
**PDF ref:** p. 3

**Source example:** 915 × 8,760 × 0.70 = 5,606,820 MWh/year. Convert MWh to kWh by multiplying by 1,000.

**Fuel required (kg/year) = annual energy output (kWh/year) / net energy yield (kWh/kg)**
Calculates fuel mass from net energy yield.

**Symbols:** Energy output in kWh/year; net yield in kWh/kg.
**Conditions:** Keep units consistent.
**PDF ref:** p. 3

**Fuel energy input = electrical energy output / thermal efficiency**
Finds required input energy.
**Symbols:** Thermal efficiency as a fraction, e.g. 0.35 for 35%.
**Conditions:** Use matching energy units.
**PDF ref:** p. 3

**Fuel mass = fuel energy input / calorific value**
Finds fuel mass from energy input.
**Symbols:** Calorific value in energy per unit mass, e.g. kWh/kg.
**Conditions:** Convert MWh to kWh if needed.
**PDF ref:** p. 3

### 2.2 SO■, CO■ and particulate emissions

**SO■ mass = sulphur mass × (64/32) = 2 × sulphur mass**
Calculates theoretical SO■ from fuel sulphur.
**Symbols:** 64 = molecular mass SO■; 32 = atomic mass S.
**Conditions:** Assumes full conversion to SO■; account for retention/removal if given.
**PDF ref:** pp. 3, 5

**CO■ mass = carbon mass × (44/12) ≈ 3.667 × carbon mass**
Calculates CO■ from carbon under complete combustion.
**Symbols:** 44 = molecular mass CO■; 12 = atomic mass C.
**Conditions:** Assumes complete conversion of carbon to CO■.
**PDF ref:** pp. 3, 5

**Carbon mass = fuel mass × (carbon percentage/100)**
Converts carbon mass percentage to carbon mass.
**Symbols:** Fuel and carbon mass use the same units.
**Conditions:** Use percentage divided by 100.
**PDF ref:** p. 3

**Uncontrolled particulate = fuel throughput × particulate generation factor**
Estimates emissions before control.
**Symbols:** Use the factor's stated units.
**Conditions:** Units must match the required output.
**PDF ref:** p. 3

**Controlled emissions = uncontrolled emissions × (1 − collection efficiency)**
Finds emissions remaining after collection.
**Symbols:** Collection efficiency as a fraction; 99.5% = 0.995.
**Conditions:** For 99.5% efficiency, residual fraction = 0.005.
**PDF ref:** p. 3

## 3. Concentration Units and Chemical Calculations

**Weight % = (mass of solute / mass of solution) × 100**
Calculates mass percentage.
**Symbols:** Use the same mass units in numerator and denominator.
**Conditions:** Denominator is total solution mass.
**PDF ref:** p. 4

**ppm by mass = (mass of solute / mass of solution) × 10■**
Expresses a mass fraction in parts per million.
**Symbols:** Mass units must match.
**Conditions:** For dilute aqueous solutions, 1 ppm ≈ 1 mg/L only when density is close to 1 kg/L.
**PDF ref:** p. 4

**Molarity (M) = moles of solute / solution volume (L)**
Calculates molar concentration.
**Symbols:** M in mol/L.
**Conditions:** Use final solution volume in litres.
**PDF ref:** p. 4

**Moles = mass (g) / molar mass (g/mol)**
Converts mass to amount in moles.
**Symbols:** Mass in g; molar mass in g/mol.
**Conditions:** Use the substance's correct molar mass.
**PDF ref:** p. 4

**x■ = moles of component i / total moles; Σx■ = 1**
Calculates and checks mole fractions.
**Symbols:** x■ = mole fraction of component i; Σ = sum over all components.
**Conditions:** For mass-based mixtures, convert each component mass to moles first. Gas volume fractions ≈ mole fractions at the same T and P.
**PDF ref:** p. 4

**Equivalents = mass (g) / equivalent weight (g/equivalent)**
Converts mass to equivalents.
**Symbols:** Equivalent weight depends on the reaction.
**Conditions:** Choose the reaction-specific equivalent weight.
**PDF ref:** p. 4

**Normality (N) = equivalents / solution volume (L)**
Expresses concentration in equivalents per litre.
**Symbols:** N in eq/L.
**Conditions:** Equivalent weight depends on the reaction.
**PDF ref:** p. 4

**Equivalent weight = molar mass / n-factor**
Computes equivalent weight when the reaction n-factor is known.
**Symbols:** n-factor = reaction-specific equivalents per mole.
**Conditions:** The source describes n-factor through H■, OH■ or charge equivalents involved.
**PDF ref:** p. 4

**Stoichiometry workflow:** balance equation → convert known mass to moles → use mole ratio → convert required moles to mass/volume/concentration → check units and significant figures (p. 4).

> **Note:** The alum/fluoride treatment example has unclear handwritten quantities and some coefficients; no unsupported reagent total is included.

## 4. Gas Mixtures and Combustion

**M■ = Σ(y■ × M■)**
Calculates average molecular weight of a gas mixture.
**Symbols:** y■ = mole/volume fraction; M■ = molecular weight of component i.
**Conditions:** Convert volume percentages to fractions; fractions should sum to 1.
**PDF ref:** p. 5

**Source example form:** M■ = yH■(2) + yO■(32) + yN■(28). The exact mixture fractions are not legible enough to verify a final answer (p. 5).

**C + O■ → CO■; 12 kg C → 44 kg CO■**
Mass ratio for complete carbon oxidation.
**Symbols:** CO■:C = 44:12.
**Conditions:** Complete combustion.
**PDF ref:** p. 5

**S + O■ → SO■; 32 kg S → 64 kg SO■**
Mass ratio for sulphur oxidation.
**Symbols:** SO■:S = 64:32.
**Conditions:** Assumes conversion to SO■.
**PDF ref:** p. 5

**2H■ + O■ → 2H■O; 4 kg H■ → 36 kg H■O**
Mass ratio for hydrogen combustion.
**Symbols:** H■O:H■ = 36:4.
**Conditions:** Use the balanced reaction shown.
**PDF ref:** p. 5

> **Note:** For incomplete combustion or excess-air problems, use the stated fuel composition, air requirement, excess-air percentage and products. Do not assume complete conversion unless stated.

## 5. Water Hardness and Equivalent Concentrations

**Equivalent weight = atomic/molecular weight ÷ ionic charge**
Finds equivalent weight for ions in hardness calculations.
**Symbols:** Use the magnitude of ionic charge in these examples.
**Conditions:** For reaction-dependent species, equivalent weight can depend on reaction.
**PDF ref:** p. 6

**Concentration as CaCO■ (mg/L) = ion concentration (mg/L) × 50 / equivalent weight**
Converts ion concentration to CaCO■-equivalent concentration.
**Symbols:** Equivalent weight of the ion.
**Conditions:** Use the ion's correct equivalent weight; report as mg/L as CaCO■.

**PDF ref:** p. 6

**Source factors:** $Ca^{2+}$: EW = 40.08/2 = 20.04; factor = 50/20.04 ≈ 2.50. $Mg^{2+}$: EW = 24.3/2 = 12.15; factor ≈ 4.12. $HCO_3^-$: EW ≈ 61; factor = 50/61 (p. 6).

**meq/L = concentration (mg/L) / equivalent weight (mg/meq)**
Converts ion concentration to milliequivalents per litre.
**Symbols:** Equivalent weight in mg/meq.
**Conditions:** Keep units consistent.
**PDF ref:** p. 6

**Hardness as $CaCO_3$ (mg/L) = meq/L × 50**
Converts meq/L to $CaCO_3$-equivalent concentration.
**Symbols:** 50 is the $CaCO_3$ factor used in the source.
**Conditions:** Report as mg/L as $CaCO_3$.
**PDF ref:** p. 6

**Ca hardness = [$Ca^{2+}$ (mg/L)] × 50/20.04**
Calculates calcium hardness.
**Symbols:** 20.04 = source's $Ca^{2+}$ equivalent weight.
**Conditions:** $Ca^{2+}$ concentration in mg/L.
**PDF ref:** p. 10

**Mg hardness = [$Mg^{2+}$ (mg/L)] × 50/12.15**
Calculates magnesium hardness.
**Symbols:** 12.15 = source's $Mg^{2+}$ equivalent weight.
**Conditions:** $Mg^{2+}$ concentration in mg/L.
**PDF ref:** p. 10

**TH = Ca hardness + Mg hardness**
Calculates total hardness.
**Symbols:** TH = total hardness; both components as $CaCO_3$.
**Conditions:** Use same units.
**PDF ref:** pp. 6, 10

**CH = min(TH, total alkalinity); NCH = TH − CH**
Estimates carbonate and non-carbonate hardness.
**Symbols:** CH = carbonate hardness; NCH = non-carbonate hardness.
**Conditions:** TH and alkalinity must be on the same $CaCO_3$-equivalent basis.
**PDF ref:** pp. 6, 10

**Source examples:** 60 mg/L $Ca^{2+}$ → ≈150 mg/L as $CaCO_3$; 29.3 mg/L $Mg^{2+}$ → ≈121 mg/L as $CaCO_3$; 366 mg/L $HCO_3^-$ → ≈300 mg/L alkalinity as $CaCO_3$ (p. 6).

## 6. BOD, COD, TOC and Theoretical Oxygen Demand

**BOD (mg/L) = (initial DO − final DO) × dilution factor**
Calculates oxygen consumed in a BOD test.
**Symbols:** DO = dissolved oxygen, typically mg/L; dilution factor is dimensionless.
**Conditions:** Source describes $BOD_3$ commonly tested for 5 days at 20 °C. Apply seed/blank correction if specified.
**PDF ref:** p. 7

**COD relationship:** COD is commonly ≥ BOD for the same sample. Source mentions potassium dichromate and about 2 hours' digestion (p. 7).

**TOC:** measures carbon in organic compounds; inorganic carbon is removed or separately accounted for. The sample may be oxidized and resulting $CO_2$ measured (p. 7).

**Note:** BOD, COD and TOC measure different properties and have no universal fixed conversion factor; the relationship depends on sample composition and biodegradability.

**$C_2H_6O_2$ + 2.5 $O_2$ → 2 $CO_2$ + 3 $H_2O$**
Complete-oxidation equation for ethylene glycol.
**Symbols:** Molar mass ≈62 g/mol; oxygen required = 2.5×32 = 80 g $O_2$/mol.
**Conditions:** Theoretical demand = (80/62) × concentration in mg/L.
**PDF ref:** pp. 7–8

**$O_2$ demand for ethylene glycol ≈ (80/62)C ≈ 1.29C mg $O_2$/L**
Calculates theoretical oxygen demand for ethylene glycol.
**Symbols:** C = ethylene glycol concentration in mg/L.
**Conditions:** Based on complete oxidation.
**PDF ref:** pp. 7–8

**$C_6H_6O$ + 7 $O_2$ → 6 $CO_2$ + 3 $H_2O$**
Complete-oxidation equation for phenol.

**Symbols:** Molar mass ≈94 g/mol; oxygen required = 7×32 = 224 g O■/mol.
**Conditions:** Theoretical demand = (224/94) × concentration in mg/L.
**PDF ref:** p. 7

**O■ demand for phenol ≈ (224/94)C ≈ 2.38C mg O■/L**
Calculates theoretical oxygen demand for phenol.
**Symbols:** C = phenol concentration in mg/L.
**Conditions:** Based on complete oxidation.
**PDF ref:** p. 7

**Source examples:** 100 mg/L ethylene glycol → ≈129 mg O■/L; 120 mg/L phenol → ≈286 mg O■/L (p. 7).

## 7. pH, Carbonate Equilibria and Alkalinity

**pH = −log■■[H■]**
Calculates pH from hydrogen-ion concentration.
**Symbols:** [H■] = hydrogen-ion concentration.
**Conditions:** Logarithm is base 10.
**PDF ref:** p. 9

**CO■(aq) + H■O ■ H■CO■* ■ H■ + HCO■■**
Represents dissolved CO■/carbonic-acid/bicarbonate equilibrium as shown in the source.
**Symbols:** H■CO■* = dissolved CO■/carbonic-acid pool in source notation.
**Conditions:** Use the appropriate equilibrium constant and stated assumptions.
**PDF ref:** p. 9

**K■ = [H■][HCO■■]/[H■CO■*]**
First dissociation equilibrium expression.
**Symbols:** Square brackets denote concentrations; K■ = first dissociation constant.
**Conditions:** Applicable constant depends on temperature and convention.
**PDF ref:** p. 9

**K■ = [H■][CO■²■]/[HCO■■]**
Second dissociation equilibrium expression.
**Symbols:** K■ = second dissociation constant.
**Conditions:** Use with equilibrium and mass-balance assumptions.
**PDF ref:** p. 9

**Alkalinity (mg/L as CaCO■) = Σ[C■ (mg/L) × 50/EW■]**
Calculates alkalinity from contributing species.
**Symbols:** C■ = concentration of species i; EW■ = equivalent weight; Σ = sum.
**Conditions:** Source lists HCO■■, CO■²■ and OH■ as principal contributors.
**PDF ref:** p. 9

**Species factors:** HCO■■: C×50/61; CO■²■: C×50/30; OH■: C×50/17. Example: 260 mg/L HCO■■ → ≈213 mg/L as CaCO■; 30 mg/L CO■²■ → 50 mg/L as CaCO■; total ≈263 mg/L as CaCO■ (p. 9).

> **Note:** pH alone does not determine a unique bicarbonate concentration. The source says temperature, equilibrium constants, total inorganic carbon and species assumptions may be required. A rainwater example gives pH ≈5.65; near pH 5.6 it treats OH■ and CO■²■ contributions as small.

## 8. Ion Balance and Sodium

**Σ cations (meq/L) ≈ Σ anions (meq/L)**
Checks ionic balance in water analysis.
**Symbols:** Convert each ion to meq/L before summing.
**Conditions:** Totals should be reasonably close; mismatch can indicate omitted ions or data errors.
**PDF ref:** p. 10

**Na■ (meq/L) = Σ anions (meq/L) − Σ known cations (meq/L)**
Finds unknown sodium from charge balance.
**Symbols:** Known cations exclude the unknown sodium; sums in meq/L.
**Conditions:** Assumes complete ion list and no important omitted ions.
**PDF ref:** p. 10

**Na■ (mg/L) = Na■ (meq/L) × 23 mg/meq**
Converts sodium to mg/L.
**Symbols:** 23 mg/meq is the source's sodium factor.
**Conditions:** Use the calculated sodium meq/L.
**PDF ref:** p. 10

## 9. Quick Constants, Conversions and Checks

| Item | Value / relationship | Use / condition |
|---|---|---|
| Molar volume at 0 °C, 1 atm | ≈22.4 L/mol | STP convention used in source |
| Kelvin conversion | K ≈ °C + 273 | Gas-volume calculations |
| Molar volume at 25 °C | ≈24.5 L/mol | From 22.4×298/273 |
| Molar volume at 100 °C | ≈30.6 L/mol | At 1 atm, from 22.4×373/273 |
| Mass conversions | 1,000 µg=1 mg; 1,000 mg=1 g; 1,000 g=1 kg | Convert before substitution |
| Metric tonne | 1 tonne=1,000 kg=10■ g | Annual emissions |
| Hours per year | 8,760 h/year | Assumes 365 days |
| Combustion ratios | CO■/C=44/12; SO■/S=64/32; H■O/H■=36/4 | Balanced reactions in source |
| Ca²■ equivalent weight | 40.08/2≈20.04 | Hardness calculation |
| Mg²■ equivalent weight | 24.3/2≈12.15 | Hardness calculation |
| CaCO■ factor | 50/EW | mg/L as CaCO■ |
| Sodium factor | 23 mg/meq | Na■ meq/L to mg/L |
| BOD test | 5 days at 20 °C | Apply stated corrections |
| COD test detail | Potassium dichromate; about 2 h digestion | As described in source |

## 10. Source Limitations

• The source reports that several handwritten vehicle counts, travel distances, emission factors, fuel-analysis values, gas-mixture fractions and alum-treatment quantities are faint or ambiguous. These are not filled in by guesswork.

• The 100 °C molar-volume value is recalculated as approximately 30.6 L/mol at 1 atm, consistent with the source’s ideal-gas temperature relationship.

• The theoretical oxygen-demand examples use the balanced equations and concentrations stated in the revised PDF; substitute the exact problem values if they differ.

• All page references refer to the 11-page *Numericals – 1: Revised Solutions* PDF: overview and quick summary on pp. 1 and 11; topic formulas on pp. 2–10.